# Panagiotis Zachos


Senior Audio & Distributed Sensing Engineer · AI Researcher

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## Profile

Applied AI/ML engineer with a strong signal processing background, building detection, localization (DoAE), and classification systems over large sensor arrays and Φ-OTDR DAS. Experienced in Python (NumPy, Pandas, PyTorch, scikit-learn), designing data pipelines, and delivering real-time analytics and visualization for geospatial and time-series data. Seeking roles in AI/ML or Data Engineering where model development, data processing, and production-ready tooling intersect.

## Education

**University of Patras** *Patras, GR*

PhD in Signal Processing *Sep 2020 – Nov 2024*

- Thesis Subject: Sound Perception Optimization for Headphone Listening
- Summary: Development and optimization of algorithms that utilize auditory mechanisms and advanced digital processing methods aiming to achieve improved quality and naturalness in the perception of sound.

**University of Patras** *Patras, GR*

Integrated Master in Electrical and Computer Engineering *Sep 2014 – Aug 2020*

- Thesis: Sound Source Localization Using Deep Neural Networks Utilizing Binaural Cues
- Grade Point Average: **7.45 / 10**

**4th General High School** *Larisa, GR*

High School *Sep 2011 – Jun 2014*

- Graduated with Distinction
- Grade Point Average: **18.8 / 20**

## Work Experience

**Satways Ltd.** *Athens, Greece*

Senior Audio & Distributed Acoustic Sensing Engineer *Sep 2025 – Present*

- Lead development of detection, DoAE, and classification models on Φ-OTDR and multi-sensor arrays; design real-time signal processing pipelines.
- Orchestrate data collection and benchmarking experiments with microphone, hydrophone, and geophone arrays alongside Φ-OTDR DAS fibers.
- Build interactive geospatial dashboards (AIS, GNSS) to plan and analyze large-scale field trials.

**Satways Ltd.** *Athens, Greece*

Audio & Distributed Acoustic Sensing Engineer *Dec 2023 – Aug 2025*

- Engineered sensor arrays in multiple configurations and geometries for long-range surveillance; standardized data capture and labeling workflows.
- Delivered robust land/sea event detection and classification across Φ-OTDR DAS and acoustic arrays, combining classical DSP with modern AI.

**ITHACA Horizon Europe Project** *Remote*

Trustworthy AI & ML Researcher *Jun 2023 – Dec 2025*

- Design systems that ensure trustworthy and ethical AI operation (security, fairness, privacy) with measurable evaluation criteria.
- Build analysis and visualization tooling for AI cybersecurity, fairness, and privacy to support model audits and reporting.
- Create gamification and argument-visualization modules to drive data-driven civic engagement.

**Sonova Holding AG** *Zurich, Switzerland*

Internship in Research & Development for Hearing Performance *Nov 2020 – Apr 2021*

- Prototyped and implemented time-domain signal-processing features in a new architecture.
- Conducted acoustic measurements.
- Built internal tooling for rapid integration and verification.
- Tuned and parameterized existing signal-processing functionalities.
- Ran hearing-performance experiments and validation.

**PROMETHEUS Project, University of Patras** *Patras, Greece*

Machine Learning Researcher for Audio-Visual Signal Processing *Sep 2020 – Dec 2023*

- Implemented a subsystem that generates static and dynamic 3D AR objects with interactive interfaces for AR-enhanced theatrical plays.
- Built a YOLO-based human pose-estimation module to track actors' movements in real time; optimized data throughput for low-latency inference.
- Designed an intuitive GUI for directing and calibrating performances; streamlined data flows for end-to-end operation.

## Selected Projects

**YouTube Album Downloader** *Github link*
Personal *Nov 2019*
- Command line application in Python to easily download full albums from YouTube and automatically split them in their respective songs by parsing the video description for song timestamps
- **Technical Skills:** Python, Shell, APIs.

**Adaptive Filtering for Headphone Active Noise Control (ANC)** *Github link*
University of Patras *Jun 2019*
- Design of 2 ANC applications utilizing adaptive filters, one with knowledge of environmental noise, via a microphone placed outside the headphone enclosure and one without information about the noise.
- **Technical Skills:** Signal Processing, MATLAB

**Blood Pressure Monitoring System** *Github link*
University of Patras *Jun 2018*
- An application written in Java consisting of two parts, a Blood Pressure Monitoring device (Raspberry Pi) with the role of the client, and a multi-threaded Server simulating a doctor's computer. The client automatically submits a patients' measurements to the server from where the doctor can remotely monitor their health.
- **Technical Skills:** Java, Raspberry Pi, Networking, Parallel/Threaded Programming.

**Intelligent Control of a ZEB with FCMs** *Patras, Greece*
University of Patras *Jun 2018*
- Design and modeling of a Zero Energy Building's power usage and production systems.
- Fuzzy Control Maps were employed to calculate the optimal energy balance to minimize carbon footprint.
- **Technical Skills:** MATLAB, Fuzzy Control Maps.

**Control System and PIDF Design** *Patras, Greece*
University of Patras *Feb 2017*
- Implementation of a transfer function (Plant) and a PIDF controller using LM741 Op-Amps.
- Performed simulations of the electrical circuits with PSpice and MATLAB Simulink to select the necessary components, laboratory measurements on the constructed circuit, and PCB designs via EAGLE Pcb for the Plant and PIDF Controller.
- **Technical Skills:** PIDF design, PCB design, Circuit simulations.

**2048 Game Solver** *Github link*
University of Patras *May 2015*
- Shell program written in C that automatically plays the popular mobile game "2048", deciding the best possible move through statistical analysis.
- **Technical Skills:** C, Statistical Analysis, Shell Game Development.

## Skills

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| **AI/ML** | Supervised/unsupervised learning, PyTorch, scikit-learn, model evaluation, feature engineering, computer vision (YOLO) |
| **Data Eng.** | Data pipelines, time-series processing, Pandas, NumPy, geospatial data (AIS, GNSS), visualization |
| **Programming** | Python (NumPy, Pandas, SciPy, PyTorch, TensorFlow, matplotlib), MATLAB, Simulink, C/C++, Java, git |
| **Platforms** | Linux (Fedora, Debian, Ubuntu), Windows |
| **Domain** | Digital Signal Processing, Distributed Acoustic Sensing (DAS), $\Phi$-OTDR, Phased Arrays, SDR, audio measurement |

## Honors & Awards

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| 2018 | **2nd Place - EESTEC Challenge**, EESTEC Challenge is an annual competition for engineering students with varying subjects. This competition involved processing Big Data and constructing Machine Learning models for planetary data. | *Greece* |
| 2013 | **Euroscola Program**, Accepted into the Euroscola program by winning the local round which involved writing an essay about the democratic deficit of the EU. During the program I was responsible for presenting a new ecological plan, devised by our team consisting of members from 5 different EU countries, in the European Parliament. | *France* |
| 2006 | **1st Place - 4th Panhellenic Orchestra Competition**, I was a member of the classical guitar ensemble from the 'Gkouletsou' conservatory, winning first place in the competition that took place in Piraeus University in Athens. | *Greece* |

## Publications

### Journal Articles

Effectiveness of L2 Regularization in Privacy-Preserving Machine Learning
N. Chandrinos, I. Loi, P. Zachos, I. Symeonidis, A. Spiliotis, M. Panou, K. Moustakas
*Under Review* (2025). 2025

A multidisciplinary analysis of transparent AI-driven toxicity detection tools for civic engagement platforms
M. Zangl, I. Loi, P. Zachos, M. Bedek, E. Dimogerontakis, C.E. Nikolaou, D. Albert, K. Moustakas
*AI & Society* (2025) pp. 1–18. 2025

Targeted Beamforming Active Noise Control Based on Disturbance Metrics
Panagiotis Zachos, Georgios Moiragias, John Mourjopoulos
*Acta Acustica* (2024). 2024

Feedforward Headphone Active Noise Control Utilizing Auditory Masking
Panagiotis Zachos, Gavriil Kamaris, John Mourjopoulos
(2023). 2023

Low Filter Order Digital Equalization for Mobile Device Earphones
Gavriil Kamaris, Panagiotis Zachos, John Mourjopoulos
*journal of the audio engineering society* 69.5 (May 2021) pp. 297–308. 2021

### Conference Proceedings

Beamforming Headphone ANC for Targeted Noise Attenuation
Panagiotis Zachos, John Mourjopoulos
*Audio Engineering Society Convention 154*, 2023, Helsinki, Finland

Targeted Beamforming ANC Based on Disturbance Metrics
Panagiotis Zachos, John Mourjopoulos
*Forum Acusticum*, 2023, Turin, Italy

Evaluation of Multichannel Reproduction of 3D audio in a Cave Automatic Virtual Environment
P Chatziantoniou, G Kamaris, K Kaleris, G Moiragias, P Zachos, S Agorgianitis, J Mourjopoulos
*ACOUSTICS 2022*, 2022, Thessaloniki, Greece

Binaural Data Reduction for Robust Sound Source Localization utilizing Convolutional Neural Networks for Reverberant Environments
Panagiotis Zachos, Gavriil Kamaris, John Mourjopoulos
*ACOUSTICS 2022*, 2022, Thessaloniki, Greece

Large Scale Headphone Response Analysis and Evaluation
Panagiotis Zachos, Dimitris Mermigas, John Mourjopoulos
*ACOUSTICS 2022*, 2022, Thessaloniki, Greece

An Efficient Data Reduction Method for Binaural Parameter Utilization
Panagiotis Zachos, Gavriil Kamaris, John Mourjopoulos
*Forum Acusticum*, 2020, Lyon, France

## Languages

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| **English** | C2 - Advanced proficiency |
| **Spanish** | B2 - Professional proficiency |
| **German** | A2 - Basic proficiency |
| **Greek** | Native proficiency |